MILITARY SPECIFICATION SHEET

PIN, SWAGE-LOCKING, OVAL 60° COUNTERSUNK HEAD, TAPERED-SECTION SHANK, MULTIPLE LOCKING GROOVES.
ALUMINUM ALLOY, CORROSION RESISTANT STEEL AND CARBON STEEL.

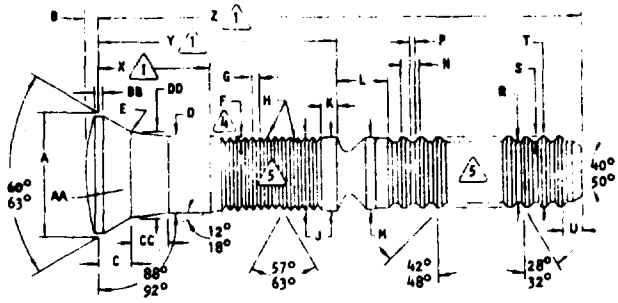
THE COMPLETE REQUIREMENTS FOR PROCURING THE SWAGE-LOCKING PINS DESCRIBED HEREIN SHALL CONSIST OF THIS DOCUMENT AND THE ISSUE IN EFFECT OF SPECIFICATION MIL-P-23469.

OVAL 60° COUNTERSUNK HEAD, SWAGE-LOCKING PINS SHALL BE FURNISHED IN THE FOLLOWING TYPE AND CLASSES. PINS MANUFACTURED TO THIS SPECIFICATION ARE INTENDED FOR USE WITH COLLARS MANUFACTURED TO MIL-P-23469.

TYPE II MULTIPLE LOCKING GROOVES (SEE FIGURE 1)

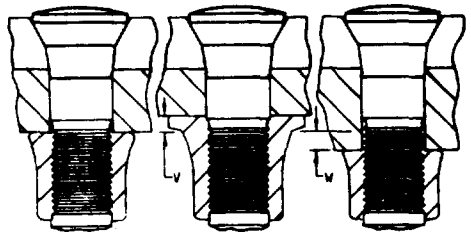
CLASS 1 ALUMINUM ALLOY
CLASS 2 CORROSION RESISTANT STEEL
CLASS 3 CARBON STEEL
16 CLASS 4 CORROSION RESISTANT STEEL - CADMIUM COATED

TYPE II - MULITPLE LOCKING GROOVE PINS



SWAGE-LOCKING FASTENER INSTALLATIONS - WHICH SHOW HOW THE TYPE OF MATING COLLAR AFFECTS GRIP RANGE OF A PIN

A



INSTALLED WITH REGULAR COLLAR | INSTALLED WITH FLANGED COLLAR | INSTALLED WITH LOW PROFILE COLLAR

NOM DIA.	A DIA. 2	AA 2	AAA 3	B 2	BB MAX.	C 2	CC	D DIA.	DD DIA.	E RAD.	F	G PITCH	H RAD.	J DIA.	K MAX.	L MAX.
1/2	.808 .768	.990 1.010	.030	.078 .093	.031	.218 .238	.219 .281	.510 .493	.530 .520	.020 .040	.025 .028	.039 .045	.002 .010	.495 .485	.094	.375
5/8	1.005 .965	1.115 1.135	.040	.102 .117	.047	.274 .294	.281 .344	.636 .617	.650 .640	.025 .050	.031 .035	.052 .058	.003 .013	.618 .608	.115	.375
3/4	1.203 1.163	1.365 1.385	.047	.125 .140	.062	.337 .357	.344 .406	.780 .741	.780 .770	.030 .060	.036 .040	.059 .065	.003 .015	.740 .730	.135	.500
7/8	1.400 1.360	1.615 1.635	.055	.149 .164	.078	.393 .413	.406 .469	.905 .866	.905 .895	.035 .065	.043 .048	.068 .076	.003 .017	.870 .860	.156	.625
1"	1.597 1.557	1.803 1.823	.062	.172 .187	.094	.454 .474	.469 .531	1.010 .990	1.030 1.020	.040 .070	.052 .058	.080 .088	.004 .020	.993 .983	.180	.625

NOM DIA.	M DIA.	N PITCH	P	R DIA.	S	T MAX. DIA.	U MIN.	V 7 FLANGED COLLAR	W 8 LOW PROFILE COLLAR	HOLE PREPARATION DATA		
										SIZE DRILL	MAX. HOLE DIA.	C'SINK DIA.
1/2	A .473 .449	.0615 .0635	.016 .018	.427 .423	.020 .023	.473	.062	.125	.140	1/2 (.500)	33/64	23/32
5/8	.615 .570	.0771 .0791	.020 .023	.530 .524	.026 .030	.590	.078	.156	.180	5/8 (.625)	41/64	15/16
3/4	.740 .682	.0927 .0947	.025 .028	.637 .631	.032 .037	.711	.094	.188	.215	3/4 (.750)	49/64	1 1/8
7/8	.860 .790	.1084 .1104	.029 .032	.746 .738	.036 .041	.828	.109	.218	.250	7/8 (.875)	57/64	1 1/4
1"	.980 .900	.124 .126	.032 .036	.857 .849	.041 .047	.951	.125	.250	.285	1" (1.000)	1 1/64	1 7/16

DIMENSIONS IN INCHES.
SHANK MUST BE STRAIGHT WITHIN .004 TOTAL INDICATOR READING (T.I.R.) PER INCH OF LENGTH.
60°/63° ANGLE AND "B" DIAMETER TO BE CONCENTRIC WITHIN .005 TOTAL INDICATOR READING (T.I.R.).

1 "X", "Y", AND "Z" SHOWN ON TABLE II. 16 CLASS 4 PINS ARE CADMIUM COATED PER QQ-P-416 TYPE I CLASS 3. A

2 DIMENSIONS "A", "AA", "B", "C", "X", "Y" & "Z" ARE FROM THEORETICAL INTERSECTION.

3 CONICAL SURFACE OF HEAD TO "D" DIAMETER MUST BE CONCENTRIC WITHIN "AAA" TOTAL INDICATOR READING (T.I.R.).

4 INITIAL LOCKING GROOVE FORMED AT JUNCTURE OF ANGLE AND "D" BODY DIAMETER.

5 DIMENSIONS INDICATED ON LOCKING GROOVES AND PULL GROOVES ARE TYPICAL.

6 ACCUMULATED PITCH TOLERANCE NOT TO EXCEED ±.002 ON ANY TEN CONSECUTIVE PULL GROOVES.

COLLAR: REGULAR HEIGHT - USE GRIP SHOWN ON TABLE II.

7 FLANGED - ADD FLANGE THICKNESS "V" TO ACTUAL THICKNESS TO BE FASTENED TO DETERMINE "GRIP" TO BE USED IN SPECIFYING DASH NUMBER, TABLE II.

8 LOW PROFILE - SUBTRACT "W" FROM ACTUAL THICKNESS TO BE FASTENED TO DETERMINE "GRIP" TO BE USED IN SPECIFYING DASH NUMBER, TABLE II.

9 REGULAR HEIGHT COLLAR REQUIRED FOR UNDER WATER APPLICATION.

FSC 5320

A FOR CHANGES SEE PAGES 1 AND 2

NOTE: MIL-P-23469/7A PINS AND MIL-P-23469/7 PINS ARE INTERCHANGEABLE TYPE FOR TYPE, CLASS FOR CLASS AND SIZE FOR SIZE

FIGURE 1

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MILITARY SPECIFICATION SHEET

TABLE II - MULTIPLE LOCKING GROOVE PINS.

DASH NUMBER IDENTIFICATION

M23469/7 ⚠1				1/2 DIAMETER				
ALUM	CRES	CARBON STEEL	CRES CAD	GRIP RANGE		X +.016 -.031	Y ±.020	Z +.00 -.26
CLASS 1	CLASS 2	CLASS 3	CLASS 4	MIN.	MAX.			
-11612	-21612	-31612	-41612	.751	1.000	.688	1.713	3.86
-11616	-21616	-31616	-41616	1.001	1.250	.938	1.963	4.11
-11620	-21620	-31620	-41620	1.251	1.500	1.188	2.213	4.36
-11624	-21624	-31624	-41624	1.501	1.750	1.438	2.463	4.61
-11628	-21628	-31628	-41628	1.751	2.000	1.688	2.713	4.86
-11632	-21632	-31632	-41632	2.001	2.250	1.938	2.963	5.11
-11636	-21636	-31636	-41636	2.251	2.500	2.188	3.213	5.36
-11640	-21640	-31640	-41640	2.501	2.750	2.438	3.463	5.61
-11644	-21644	-31644	-41644	2.751	3.000	2.688	3.713	5.86
-11648	-21648	-31648	-41648	3.001	3.250	2.938	3.963	6.11
-11652	-21652	-31652	-41652	3.251	3.500	3.188	4.213	6.36
-11656	-21656	-31656	-41656	3.501	3.750	3.438	4.463	6.61
-11660	-21660	-31660	-41660	3.751	4.000	3.688	4.713	6.86
-11664	-21664	-31664	-41664	4.001	4.250	3.938	4.963	7.11

M23469/7 ⚠1				3/4 DIAMETER				
ALUM	CRES	CARBON STEEL	CRES CAD	GRIP RANGE		X +.016 -.031	Y ±.020	Z +.00 -.26
CLASS 1	CLASS 2	CLASS 3	CLASS 4	MIN.	MAX.			
-12416	-22416	-32416	-42416	1.001	1.250	.938	2.268	5.09
-12420	-22420	-32420	-42420	1.251	1.500	1.188	2.518	5.34
-12424	-22424	-32424	-42424	1.501	1.750	1.438	2.768	5.59
-12428	-22428	-32428	-42428	1.751	2.000	1.688	3.018	5.84
-12432	-22432	-32432	-42432	2.001	2.250	1.938	3.268	6.09
-12436	-22436	-32436	-42436	2.251	2.500	2.188	3.518	6.34
-12440	-22440	-32440	-42440	2.501	2.750	2.438	3.768	6.59
-12444	-22444	-32444	-42444	2.751	3.000	2.688	4.018	6.84
-12448	-22448	-32448	-42448	3.001	3.250	2.938	4.268	7.09
-12452	-22452	-32452	-42452	3.251	3.500	3.188	4.518	7.34
-12456	-22456	-32456	-42456	3.501	3.750	3.438	4.768	7.59
-12460	-22460	-32460	-42460	3.751	4.000	3.688	5.018	7.84
-12464	-22464	-32464	-42464	4.001	4.250	3.938	5.268	8.09

M23469/7 ⚠1				5/8 DIAMETER				
ALUM	CRES	CARBON STEEL	CRES CAD	GRIP RANGE		X +.016 -.031	Y Ⓐ ±.020	Z Ⓐ +.00 -.26
CLASS 1	CLASS 2		CLASS 3	MIN.	MAX.			
-12012	-22012		-42012	.751	1.000	.688	1.863	4.28
-12016	-22016		-42016	1.001	1.250	.938	2.113	4.53
-12020	-22020		-42020	1.251	1.500	1.188	2.363	4.78
-12024	-22024		-42024	1.501	1.750	1.438	2.613	5.03
-12028	-22028		-42028	1.751	2.000	1.688	2.863	5.28
-12032	-22032		-42032	2.001	2.250	1.938	3.113	5.53
-12036	-22036		-42036	2.251	2.500	2.188	3.363	5.78
-12040	-22040		-42040	2.501	2.750	2.438	3.613	6.03
-12044	-22044		-42044	2.751	3.000	2.688	3.863	6.28
-12048	-22048		-42048	3.001	3.250	2.938	4.113	6.53
-12052	-22052		-42052	3.251	3.500	3.188	4.363	6.78
-12056	-22056		-42056	3.501	3.750	3.438	4.613	7.03
-12060	-22060		-42060	3.751	4.000	3.688	4.863	7.28
-12064	-22064		-42064	4.001	4.250	3.938	5.113	7.53

M23469/7 ⚠1				7/8 DIAMETER				
ALUM	CRES	CARBON STEEL	CRES CAD	GRIP RANGE		X +.016 -.031	Y ±.020	Z +.00 -.26
CLASS 1	CLASS 2	CLASS 3	CLASS 4	MIN.	MAX.			
-12816	-22816	-32816	-42816	1.001	1.250	.938	2.421	5.38
-12820	-22820	-32820	-42820	1.251	1.500	1.188	2.671	5.63
-12824	-22824	-32824	-42824	1.501	1.750	1.438	2.921	5.88
-12828	-22828	-32828	-42828	1.751	2.000	1.688	3.171	6.13
-12832	-22832	-32832	-42832	2.001	2.250	1.938	3.421	6.38
-12836	-22836	-32836	-42836	2.251	2.500	2.188	3.671	6.63
-12840	-22840	-32840	-42840	2.501	2.750	2.438	3.921	6.88
-12844	-22844	-32844	-42844	2.751	3.000	2.688	4.171	7.13
-12848	-22848	-32848	-42848	3.001	3.250	2.938	4.421	7.38
-12852	-22852	-32852	-42852	3.251	3.500	3.188	4.671	7.63
-12856	-22856	-32856	-42856	3.501	3.750	3.438	4.921	7.88
-12860	-22860	-32860	-42860	3.751	4.000	3.688	5.171	8.13
-12864	-22864	-32864	-42864	4.001	4.250	3.938	5.421	8.38

M23469/7 ⚠1				5/8 DIAMETER				
		CARBON STEEL		GRIP RANGE		X +.016 -.031	Y ±.020	Z +.00 -.26
		CLASS 3		MIN.	MAX.			
		-32012		.751	1.000	.688	1.933	4.34
		-32016		1.001	1.250	.938	2.183	4.59
		-32020		1.251	1.500	1.188	2.433	4.84
		-32024		1.501	1.750	1.438	2.683	5.09
		-32028		1.751	2.000	1.688	2.933	5.34
		-32032		2.001	2.250	1.938	3.183	5.59
		-32036		2.251	2.500	2.188	3.433	5.84
		-32040		2.501	2.750	2.438	3.683	6.09
		-32044		2.751	3.000	2.688	3.933	6.34
		-32048		3.001	3.250	2.938	4.183	6.59
		-32052		3.251	3.500	3.188	4.433	6.84
		-32056		3.501	3.750	3.438	4.683	7.09
		-32060		3.751	4.000	3.688	4.933	7.34
		-32064		4.001	4.250	3.938	5.183	7.59

M23469/7 ⚠1				1" DIAMETER				
ALUM	CRES	CARBON STEEL	CRES CAD	GRIP RANGE		X +.016 -.031	Y ±.020	Z +.00 -.26
CLASS 1	CLASS 2	CLASS 3	CLASS 4	MIN.	MAX.			
-13220	-23220	-33220	-43220	1.251	1.500	1.188	2.823	6.32
-13224	-23224	-33224	-43224	1.501	1.750	1.438	3.073	6.57
-13228	-23228	-33228	-43228	1.751	2.000	1.688	3.323	6.82
-13232	-23232	-33232	-43232	2.001	2.250	1.938	3.573	7.07
-13236	-23236	-33236	-43236	2.251	2.500	2.188	3.823	7.32
-13240	-23240	-33240	-43240	2.501	2.750	2.438	4.073	7.57
-13244	-23244	-33244	-43244	2.751	3.000	2.688	4.323	7.82
-13248	-23248	-33248	-43248	3.001	3.250	2.938	4.573	8.07
-13252	-23252	-33252	-43252	3.251	3.500	3.188	4.823	8.32
-13256	-23256	-33256	-43256	3.501	3.750	3.438	5.073	8.57
-13260	-23260	-33260	-43260	3.751	4.000	3.688	5.323	8.82
-13264	-23264	-33264	-43264	4.001	4.250	3.938	5.573	9.07

⚠1 PART NUMBER EXAMPLE - M23469/7-11612

BASIC OR GENERAL SPECIFICATION
STYLE - OVAL 60° COUNTERSUNK HEAD
GRIP IN 1/16" (7/8 =1/8)
DIAMETER IN 1/32" (1/2)
MATERIAL - ALUMINUM ALLOY

DIMENSIONS IN INCHES

Ⓐ FOR CHANGES SEE PAGES 1 AND 2

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